

College Physics II

Lecture summaries

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Abstract

Disclaimer: These notes *summarize* the key concepts and formulae discussed during lecture. They should be supplemented by any standard algebra-based physics textbook on these topics (e.g., by Rex and Wolfson) or internet resource to fill in any gaps. Please send any comments, criticisms, suggestions to: joseph.d.romano@gmail.com.

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1 Electric charges, forces, and fields

1.1 Electric charge

- Properties of electric charge:
 - (i) Two types: *positive* and *negative* (+ and −), whose effects tend to cancel. Opposite charges attract, like charges repel.
 - (ii) Total electric charge is *conserved* (cannot be created or destroyed, only transferred from one object to another).
 - (iii) Electric charge is *quantized* (integer multiples of the charge of a proton $e = 1.602 \times 10^{-19}$ C; an electron has charge $-e$).
- Materials:
 - (i) *Conductors*: one or more electrons per atom are able to move freely through the material. Examples: aluminum, copper, gold, ... (metals).
 - (ii) *Insulators*: electrons are bound to atomic nuclei. Examples: rubber, wood, paper, ...
- Some demos:
 - 1) A plastic rod rubbed with fur is negatively charged; a glass rod rubbed with silk is positively charged. Show that opposite charges attract, like charges repel.
 - 2) Bits of uncharged paper will be attracted to both negative and positive rods due to charge separation within the atoms / molecules in the paper (an insulator).
 - 3) A charged rubber balloon sticks to the white board for the same reason as (2).

1.2 Electric force

- *Coulomb's law*: The magnitude of the electric force between two point charges q_1 and q_2 separated by a distance r is

$$F = \frac{k|q_1||q_2|}{r^2} \quad \text{where} \quad k = 8.99 \times 10^9 \frac{\text{N} \cdot \text{m}^2}{\text{C}^2} \quad (1)$$

- The force is attractive (directed towards one another) if the two charges have opposite signs; the force is repulsive (directed away from one another) if the two charges have the same signs.
- One sometimes writes

$$k = \frac{1}{4\pi\epsilon_0}, \quad \text{where} \quad \epsilon_0 = 8.85 \times 10^{-12} \frac{\text{C}^2}{\text{N} \cdot \text{m}^2} \quad (2)$$

where ϵ_0 is the so-called *permittivity* of free space (vacuum).

- The electric force on point charge Q due to a set of point charges $q_1, q_2, \dots, q_N$ is given by the vector sum of the forces

$$\vec{F}_{Q,\text{net}} = \vec{F}_{1Q} + \vec{F}_{2Q} + \dots + \vec{F}_{NQ} \quad (3)$$

where $\vec{F}_{1Q}$ is the electric force that point charge q_1 exerts on Q , etc. This is called the *superposition principle*.

1.3 Electric field

- The electric force between charges is a *non-contact* force. A set of charges creates a *field* to which another charge responds.

- The electric field at a point P in space is a vector $\vec{E}(P)$, whose direction is given by the electric force $\vec{F}_{q_0}(P)$ on a small, positive (so called “test”) charge q_0 placed at P , and whose magnitude is given by $F_{q_0}(P)/q_0$:

$$\vec{E}(P) = \vec{F}_{q_0}(P)/q_0 \quad (4)$$

Note that by dividing by q_0 , the electric field does not depend on properties of the test charge. It depends only on the charges producing the field.

- For a single point charge q :

$$E(P) = \frac{kq}{r^2} \quad (5)$$

where point P is a distance r from q . The direction of the electric field is radially away from the charge if $q > 0$ and towards the charge if $q < 0$.

- The electric field produced by a set of discrete point charges $q_1, q_2, \dots, q_N$ is given by the superposition principle:

$$\vec{E}(P) = \vec{E}_1(P) + \vec{E}_2(P) + \dots + \vec{E}_N(P) \quad (6)$$

where $\vec{E}_1(P)$ is the electric field due to q_1 , etc.

1.4 Electric field examples

- Electric dipole: Two charges, $+q$ and $-q$ separated by distance a along the x -axis (located at $x = -a/2$, $x = a/2$, respectively). Then the electric field at point $P = (0, y)$ on the y -axis is given by:

$$E_y(P) = 0, \quad E_x(P) = \frac{kqa}{[y^2 + (a/2)^2]^{3/2}} \quad (7)$$

When $y \gg a$, the electric field has magnitude $E(P) \simeq kqa/y^3$, which falls off as the *cube* of the distance to the dipole.

- The electric field produced by an infinite 2-dimensional sheet of charge with surface charge density σ has magnitude

$$E(P) = \frac{|\sigma|}{2\epsilon_0} \quad (8)$$

The field points perpendicular to the sheet (away from the sheet if $\sigma > 0$; toward the sheet if $\sigma < 0$). The magnitude of the electric field is independent of the distance from the sheet.

- In practice, one doesn't have 2-dimensional sheets that are infinite in extent. But the above results still hold for a finite sheet provided the point P is sufficiently close to the sheet.
- The electric field between two infinite, parallel sheets of charge with surface charge densities $+\sigma$ and $-\sigma$ has magnitude

$$E(P) = \frac{\sigma}{\epsilon_0} \quad (9)$$

The electric field points from the positively-charged sheet to the negatively-charged sheet.

- This last example will be important when we discuss *capacitors* later on.

1.5 Electric field lines

- Pictorial representation of the electric field $\vec{E}$. Direction of field lines point in the direction of $\vec{E}$; density of field lines (number per area perpendicular to $\vec{E}$) is proportional to the magnitude of $\vec{E}$.

- Rules for drawing electric field lines:
 - (i) lines begin on $+$ charges, end on $-$ charges (or may extend to infinity);
 - (ii) the number of lines leaving or terminating on a charge should be proportional to the magnitude of the charge;
 - (iii) electric field lines cannot cross.
- For a point charge, the density of lines at a distance r from the charge is given by $N/4\pi r^2$, which is proportional to the magnitude of the field $E(P) = kq/r^2$ as it should be.

1.6 Conductors in electrostatic fields

- Properties:
 - 1) $\vec{E} = \vec{0}$ everywhere inside a conductor, for both solid conductors and conductors with *empty* cavities.
 - 2) If an isolated conductor carries a charge, then that excess charge resides on the outer surface of the conductor.
 - 3) The electric field just outside the surface of a charged conductor is perpendicular to the surface with magnitude $E = \sigma/\epsilon_0$, where σ is the surface charge density.
 - 4) For irregularly shaped conductors, the surface charge density σ is greatest where the radius of curvature of the surface is smallest (i.e., where the surface is sharpest).
- Property (1) arises since charges in the conductor rearrange themselves in response to an applied electric field such that the *induced* field associated with these charges exactly cancels the applied field in the conductor, $\vec{E} + \vec{E}_{\text{ind}} = \vec{0}$.
- Property (1) implies that any external electric field is shielded from a cavity inside a conductor. That's why you're safe inside a car during a lightning storm.
- Fields within a cavity, produced e.g., by a charge, do make it outside of the conductor. So the outside of a conductor is not shielded from the fields inside a cavity of a conductor.